

Project Report

Import Data Using Transform Maps (Spreadsheet)

Platform	ServiceNow
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1. Introduction

1.1 Project Overview

This project shows how to bring employee training data from a spreadsheet into ServiceNow and keep it secure. An Excel (.xlsx) file is loaded into an import set table, a transform map moves the rows into a custom table called Employee Training Records, and each record is linked to an employee in the sys_user table. The employee's department is pulled into the record by dot-walking. Finally, a custom Hr Manager role and Access Control Lists (ACLs) decide who can read and write the records, and the result is tested by impersonating different users.

1.2 Purpose

The purpose of the project is to give a simple, repeatable way to load bulk data into ServiceNow without typing each record, to link every record to an employee and show employee details such as department for easier reporting, and to protect the data with role-based access control.

2. Ideation Phase

2.1 Problem Statement

Training records are kept in spreadsheets and typed into the system one by one, which is slow and error-prone. Each record also has to be linked to an employee, and details such as the employee's department must be available in the record for easier reporting. At the same time, only the right people should be able to see or change these records.

2.2 Empathy Map Canvas

Persona: an HR administrator who maintains employee training records in ServiceNow.

Says	Thinks
"I have 20 rows in a spreadsheet, I don't want to type them again." "Which employee does this training belong to?"	"Will the import create duplicates or wrong values?" "Can anyone else see these records?"
Does	Feels
Prepares the spreadsheet, uploads it, maps columns to fields, runs the transform and checks the records.	Worried about mistakes in the data and about leaking personal information; relieved when the import shows Success.
Pains	Gains
Manual data entry; mismatched column names and choice values; no department shown in the record; unclear who has access.	Fast bulk import; employee and department shown automatically; only HR managers can view and edit the data.

2.3 Brainstorming

Ideas the team considered:

- Enter each training record manually in the form.
- Import the spreadsheet directly into the target table.
- Load the spreadsheet into an import set (staging) table and use a transform map to move it into the target table.

- Copy the department into each record with a script, or show it through dot-walking.
- Hide the data with a client script, or protect it with roles and ACLs.

Selected approach: Import Set with a Transform Map, because staged rows can be checked before they reach the real table; dot-walking, because the department stays in one place (sys_user); and roles with ACLs, because they are enforced by the platform on every request.

3. Requirement Analysis

3.1 Customer Journey Map

The journey of the administrator who imports and secures the data:

Phase	Steps	Goal	Positive	Negative	Opportunity
Prepare	Clean the spreadsheet; choose the target table and key	Avoid bad data and duplicates	A tidy sheet makes later steps smooth	Unclear key column; merged or blank headers	Keep a template sheet with exact headers
Load	Create the data source; upload the file with Load Data	Get the file in without manual entry	Headers become fields automatically; nothing touches the target yet	Wrong sheet or header row loads wrong data	Test with a few rows first
Map	Create the transform map; add field maps	Connect columns to the right fields	Auto Map handles matching names	References and choices must match exactly	Normalize values before loading
Transform	Run the transform	Run once and get it right	Many rows moved in one click	Vague errors; reruns can repeat rows	Use coalesce to update, not duplicate
Verify	Check records; dot-walk department; set roles and ACLs; test by impersonation	Trust the result and protect the data	Row status is clear; ACLs enforce access	Easy to forget the ACL on the table	Test every role before release

3.2 Solution Requirement

Functional requirements

FR No.	Requirement	Sub-requirement
FR-1	Table setup	Create Employee Training Records with Training Name, Completion Date, Status (choice) and Employee (reference to sys_user)
FR-2	Data import	Load an .xlsx file with Load Data into an import set table
FR-3	Transformation	Create a transform map and field maps, then run the transform
FR-4	Employee details	Show Employee.Department on the form and list by dot-walking
FR-5	Roles and users	Create a user and the Hr Manager role; add the role to the application menu and module
FR-6	Access control	Create read and write ACLs that require the Hr Manager role
FR-7	Testing	Verify access by impersonating an HR manager and another user

Non-functional requirements

NFR No.	Requirement	Description
NFR-1	Usability	Uses standard ServiceNow forms and lists; no coding is needed
NFR-2	Security	Only users with the Hr Manager role can read or write the table
NFR-3	Reliability	Staging table lets rows be checked before they reach the target table
NFR-4	Performance	Bulk rows load in one operation; department is read through dot-walking, not copied
NFR-5	Availability	Runs on the ServiceNow-hosted cloud instance
NFR-6	Scalability	The same transform map can be reused for later imports

3.3 Data Flow Diagram

The diagrams show how data moves from the spreadsheet into ServiceNow and how access is controlled.

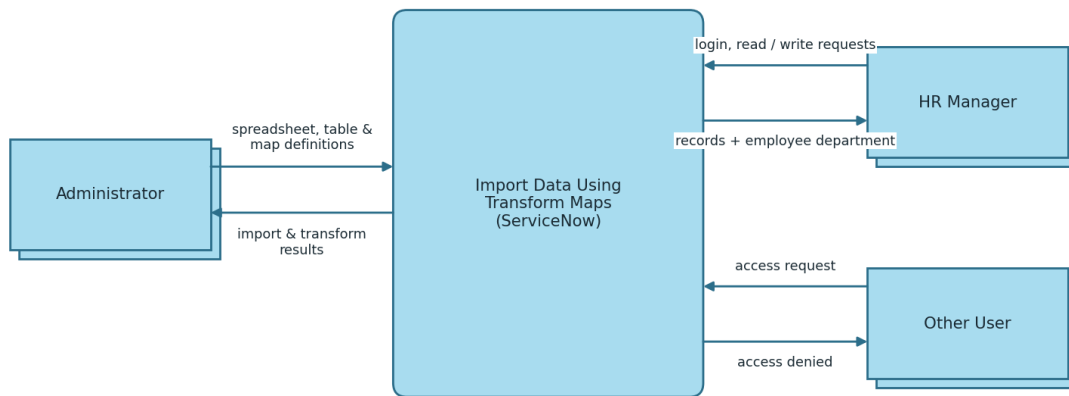


Figure 3.1: DFD Level 0

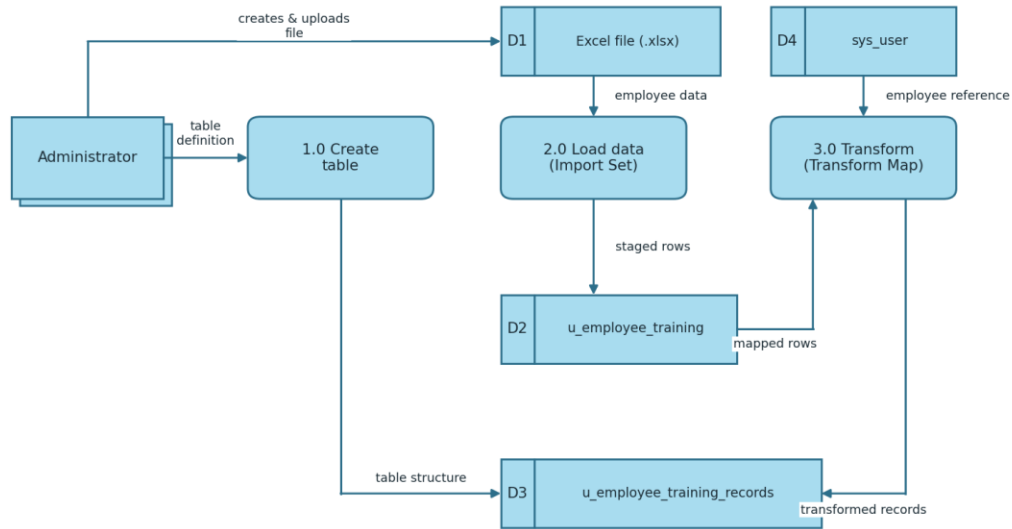


Figure 3.2: DFD Level 1 (importing data with a transform map)

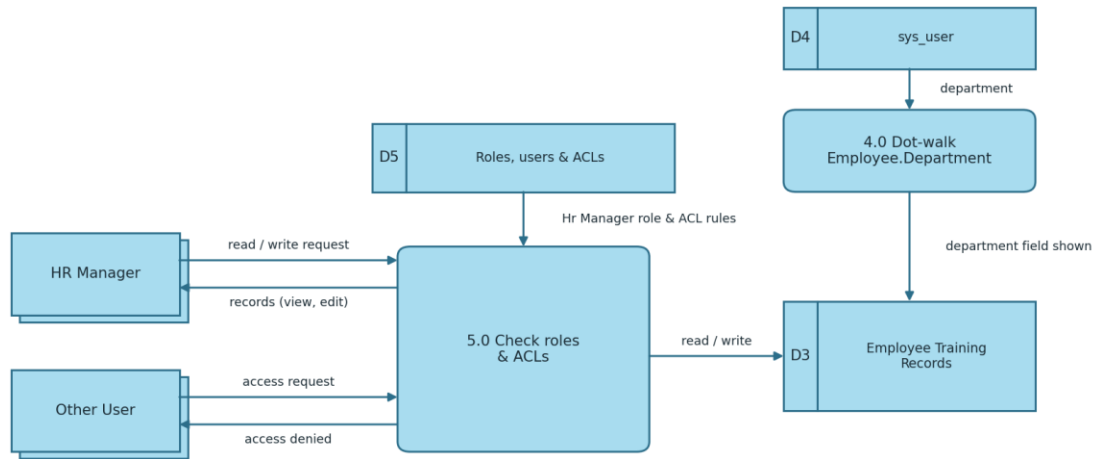


Figure 3.3: DFD Level 1 (dot-walking and access control)

3.4 Technology Stack

S.No	Component	Technology
1	User Interface	ServiceNow Next Experience (Platform UI) in a web browser
2	Application Logic-1 (Data import)	Import Sets (Load Data)
3	Application Logic-2 (Transform)	Transform Map and Field Maps
4	Application Logic-3 (Dot-walking)	Form Layout (dot-walking)
5	Application Logic-4 (Access control)	Roles, Users and Access Control (ACL)

S.No	Component	Technology
6	Database	Tables u_employee_training, u_employee_training_records and sys_user
7	Cloud Database	ServiceNow cloud instance database
8	File Storage	Google Sheets / Excel (.xlsx) uploaded through Load Data
9	External API / Machine Learning	Not used
10	Infrastructure	ServiceNow cloud instance (application scope: Global)

4. Project Design

4.1 Problem Solution Fit

Problem	Why it hurts	How the solution fits
Typing records one by one	Slow and error-prone	Bulk import from a spreadsheet with Load Data and a transform map
Records not tied to employees	Reporting is difficult	Employee reference field to sys_user
Department missing in the record	Extra lookup needed	Dot-walking shows Employee.Department
Anyone could open the data	Privacy risk	Hr Manager role with read and write ACLs

4.2 Proposed Solution

Create a custom table for training records and load the spreadsheet into an import set (staging) table. A transform map with field maps moves the rows into the custom table and links each row to an employee. The employee's department is added to the form and list by dot-walking. A new Hr Manager role is added to the application menu, the module and the read and write ACLs, so only HR managers can see and edit the records. The result is tested by impersonating an HR manager (access granted) and another user (access denied).

4.3 Solution Architecture

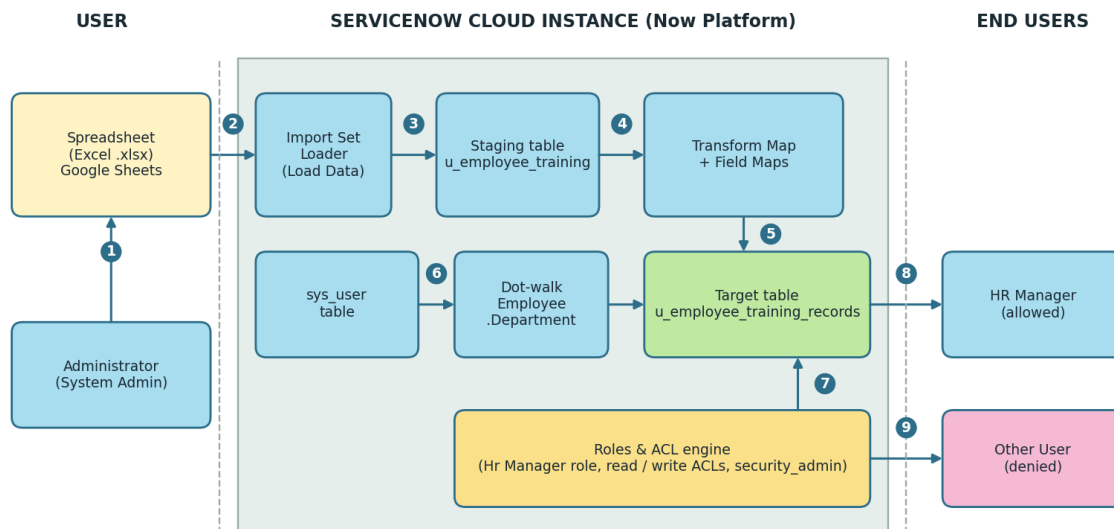


Figure 4.1: Solution architecture

The administrator uploads the spreadsheet to the ServiceNow instance. The Import Set Loader stages the rows in u_employee_training, the transform map inserts them into u_employee_training_records, and dot-walking reads the department from sys_user. The Roles and ACL engine checks every request: the HR manager is allowed and other users are denied.

5. Project Planning & Scheduling

5.1 Project Planning

Product backlog and sprint schedule

Sprint	Epic	User Story No.	User Story / Task	Story Points	Priority
Sprint-1	Table creation	USN-1	As an administrator, I can create the Employee Training Records table.	2	High
Sprint-1		USN-2	As an administrator, I can add the fields Training Name, Completion Date, Status and Employee.	2	High
Sprint-1		USN-3	As an administrator, I can add the choice values to the Status field.	1	Medium
Sprint-1	Data import	USN-4	As an administrator, I can prepare the .xlsx file.	1	High
Sprint-1		USN-5	As an administrator, I can upload the file using Load Data.	2	High
Sprint-1	Transform map	USN-6	As an administrator, I can create a transform map.	2	High
Sprint-1		USN-7	As an administrator, I can add field maps.	3	High
Sprint-1		USN-8	As an administrator, I can run the transform.	2	High
Sprint-2	Dot-walking	USN-9	As an administrator, I can dot-walk Employee.Department onto the form layout.	3	Medium
Sprint-2	Roles & users	USN-10	As an administrator, I can create a new user (backup user).	1	High
Sprint-2		USN-11	As an administrator, I can create the Hr Manager role.	1	High
Sprint-2		USN-12	As an administrator, I can assign the Hr Manager role to the user.	2	High
Sprint-2		USN-13	As an administrator, I can add the role to the application menu and module.	3	Medium
Sprint-3	ACL	USN-14	As an administrator, I can elevate my role to security_admin.	1	High
Sprint-3		USN-15	As an administrator, I can create a read ACL for the Hr Manager role.	3	High
Sprint-3		USN-16	As an administrator, I can create a write ACL for the Hr Manager role.	2	High
Sprint-3	Testing	USN-17	As an HR manager, I can view and edit the records.	3	High
Sprint-3		USN-18	As another user, I cannot see the table.	2	High

Project tracker

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	15	6 Days	05 Oct 2026	10 Oct 2026
Sprint-2	10	6 Days	12 Oct 2026	17 Oct 2026
Sprint-3	11	6 Days	19 Oct 2026	24 Oct 2026
Total	36	18 Days		

Velocity

Velocity = total story points / number of sprints = $36 / 3 = 12$ story points per sprint. Average velocity per day = $12 / 6 = 2$ story points per day.

Burndown chart

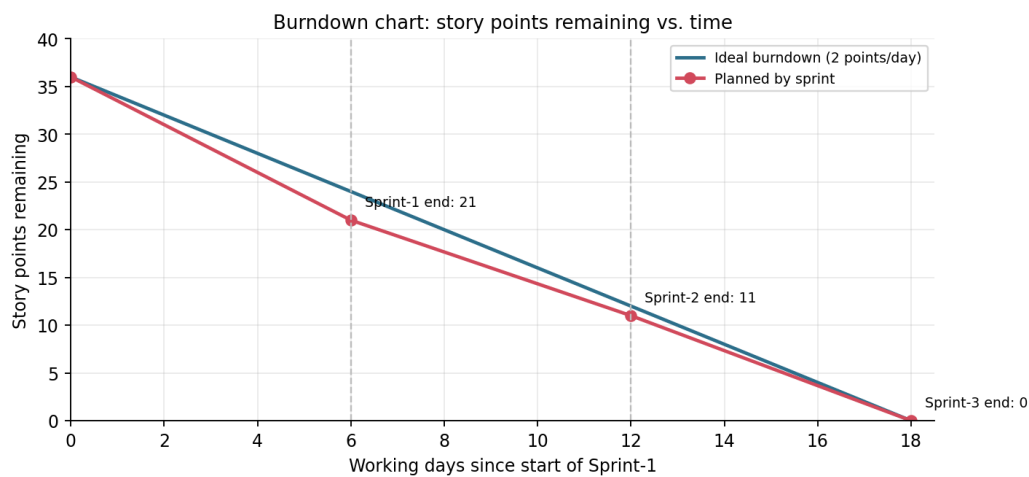


Figure 5.1: Burndown chart (ideal vs. planned by sprint)

6. Functional and Performance Testing

6.1 Performance Testing

User Acceptance Testing was carried out on each part of the project. The defect analysis shows resolved or closed bugs by severity and how they were resolved.

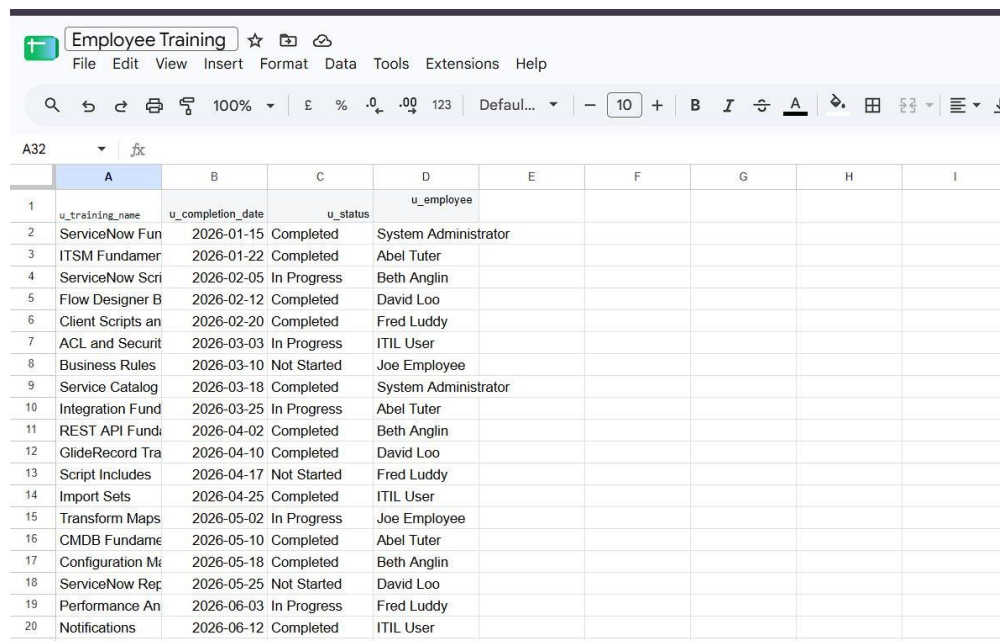
Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	0	0	0	1	1
Duplicate	0	0	0	0	0
External	0	0	0	0	0
Fixed	0	2	0	0	2
Not Reproduced	0	0	0	0	0
Skipped	0	0	0	0	0
Won't Fix	0	0	0	0	0
Totals	0	2	0	1	3

The test case analysis shows the number of test cases that passed, failed and were not tested.

Section	Total Cases	Not Tested	Fail	Pass
Table Creation	4	0	0	4
Data Import	3	0	0	3
Transform Map	4	0	0	4
Dot-Walking	2	0	0	2
Roles and Users	4	0	0	4
Security (ACL)	3	0	0	3
Result Testing (Impersonation)	3	0	0	3
Totals	23	0	0	23

7. Results

7.1 Output Screenshots



	A	B	C	D	E	F	G	H	I
1	u_training_name	u_completion_date	u_status	u_employee					
2	ServiceNow Fun	2026-01-15	Completed	System Administrator					
3	ITSM Fundamer	2026-01-22	Completed	Abel Tuter					
4	ServiceNow Scri	2026-02-05	In Progress	Beth Anglin					
5	Flow Designer B	2026-02-12	Completed	David Loo					
6	Client Scripts an	2026-02-20	Completed	Fred Luddy					
7	ACL and Securit	2026-03-03	In Progress	ITIL User					
8	Business Rules	2026-03-10	Not Started	Joe Employee					
9	Service Catalog	2026-03-18	Completed	System Administrator					
10	Integration Fund	2026-03-25	In Progress	Abel Tuter					
11	REST API Funda	2026-04-02	Completed	Beth Anglin					
12	GlideRecord Tra	2026-04-10	Completed	David Loo					
13	Script Includes	2026-04-17	Not Started	Fred Luddy					
14	Import Sets	2026-04-25	Completed	ITIL User					
15	Transform Maps	2026-05-02	In Progress	Joe Employee					
16	CMDB Fundame	2026-05-10	Completed	Abel Tuter					
17	Configuration M	2026-05-18	Completed	Beth Anglin					
18	ServiceNow Rep	2026-05-25	Not Started	David Loo					
19	Performance An	2026-06-03	In Progress	Fred Luddy					
20	Notifications	2026-06-12	Completed	ITIL User					

Figure 7.1: Employee Training spreadsheet (.xlsx) used as the data source

servicenow All Favorites History Workspaces Admin Load Data ☆ Search

Load Data Submit

Import set table ☒ Create table ☐ Existing table

* Label Employee Training

Name u_employee_training

Source of the import ☒ File ☐ Data source

File Choose File | Employee Training.xlsx

Sheet number 1

Header row 1

Submit

Figure 7.2: Load Data creates the import set table `u_employee_training`

servicenow All Favorites History Workspaces Admin ServiceNow ☆ Search

Progress

Name ImportProcessor

State Complete

Completion code Success

Message Processed: 4, inserts 3, updates 0, errors 0, empty and ignored 1, ignored errors 0 (0:00:00.915)

Next steps...

[Import sets](#) Go to the import sets for this data load

[Loaded data](#) Go to the newly imported data inside the staging table: u_employee_training

[Create transform map](#) Create a transform map for the newly staged data

[Import log](#) View the import log

Figure 7.3: Import completed with State Complete and Success

servicenow All Favorites History Workspaces Admin Table Transform Map - ... ☆ Search

Table Transform Map New record Submit

* Name Employee Transform

* Source table Employee Training [u_employ...]

Active ☒

Run business rules ☒

Enforce mandatory fields No

Copy empty fields ☐

Create new record on empty coalesce fields ☐

Application Global

Created

* Target table Employee Training Records [u...]

Order 100

Run script ☐

Submit

Figure 7.4: Transform map linking the import set table to the target table

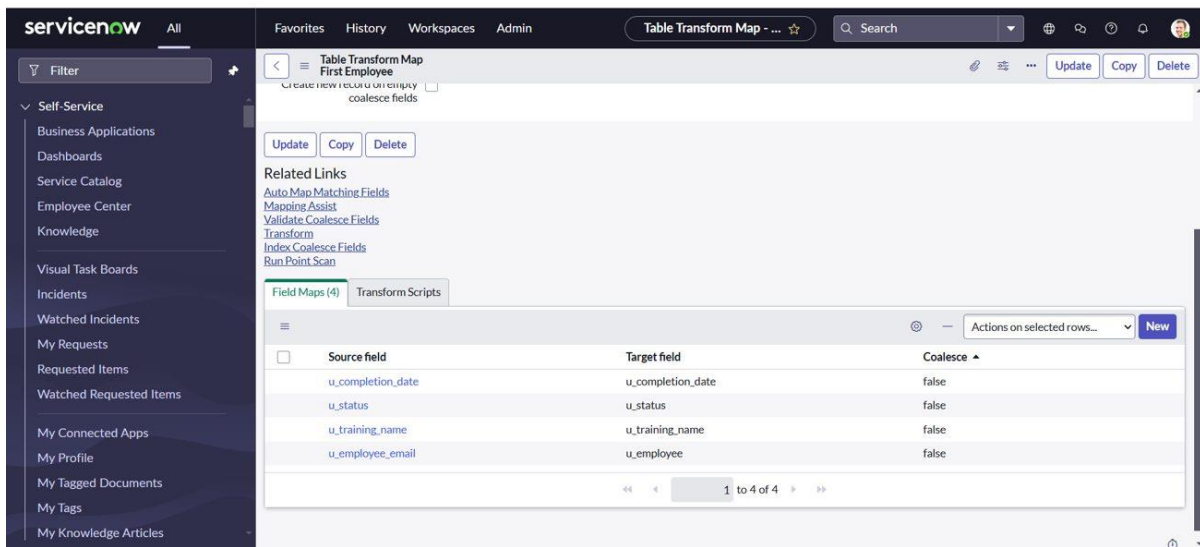


Figure 7.5: Field maps added to the transform map

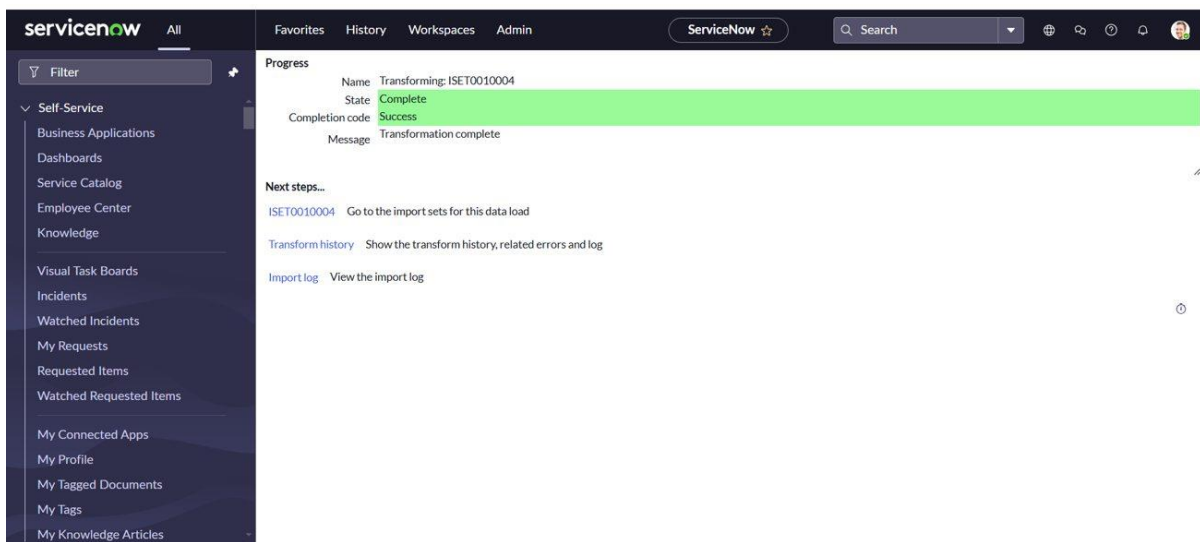


Figure 7.6: Transformation completed successfully

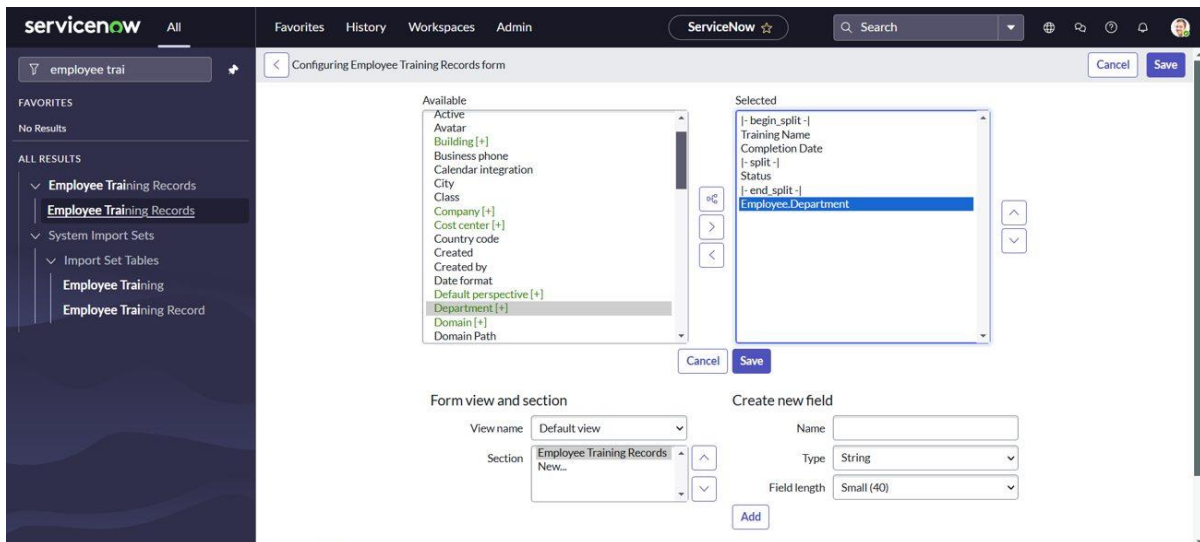


Figure 7.7: Dot-walking Employee.Department in the form layout

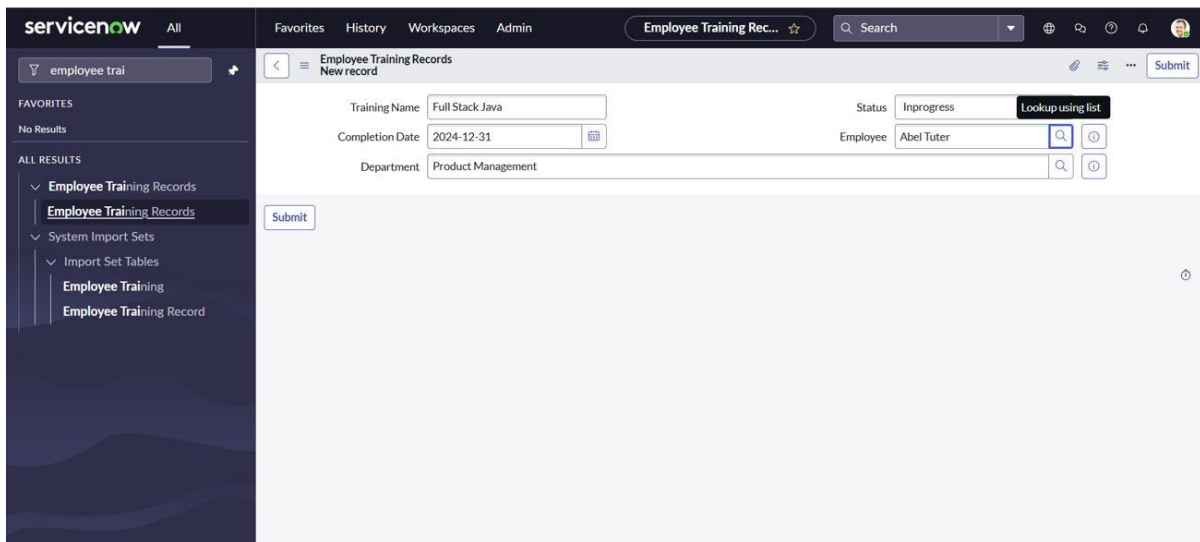


Figure 7.8: Employee Training Records form showing the Department field

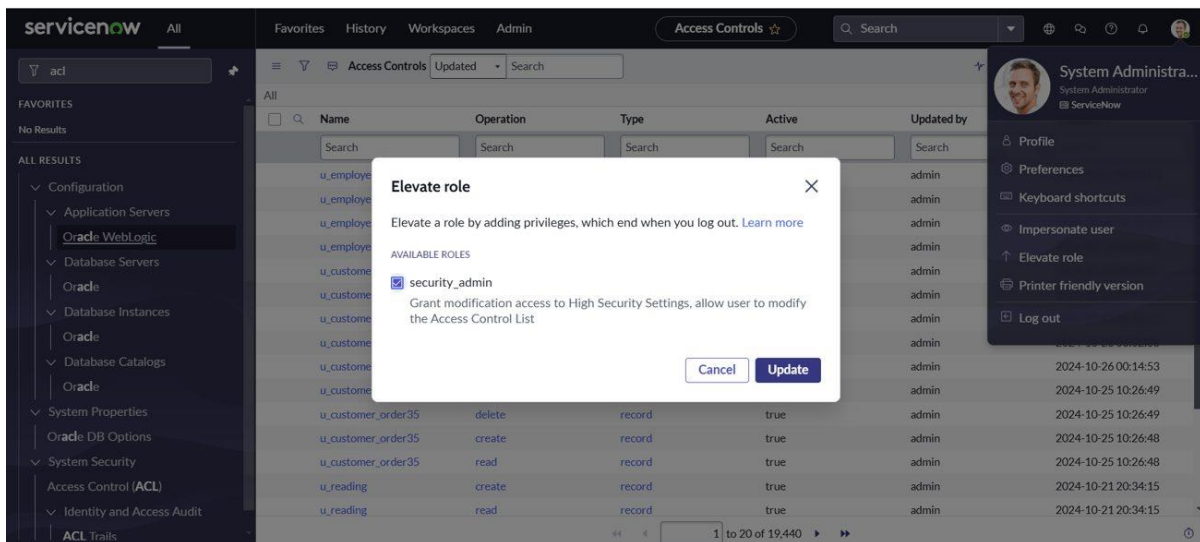


Figure 7.9: Elevating the role to security_admin

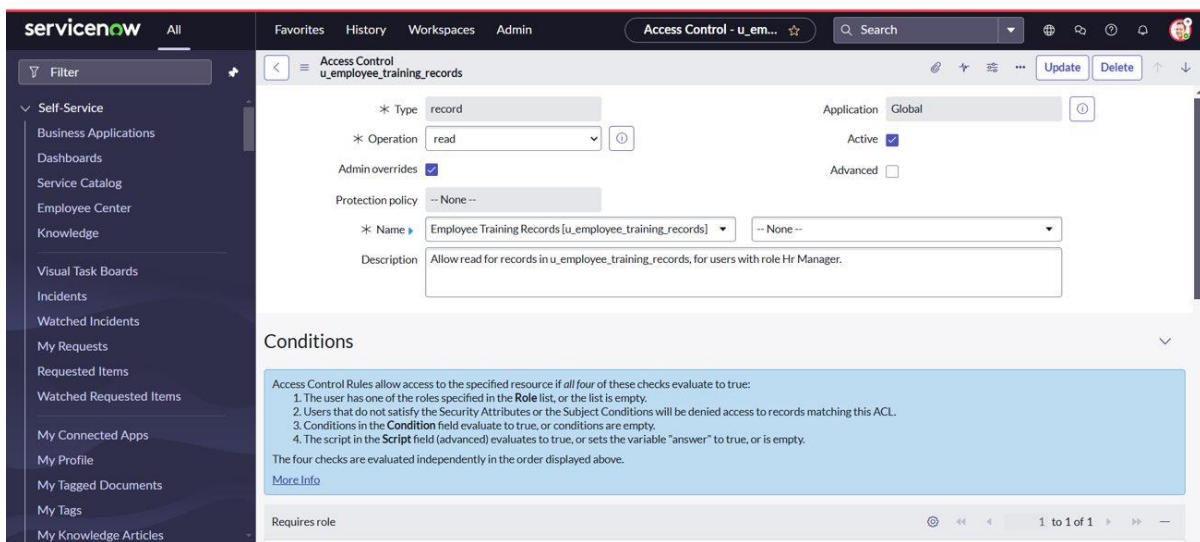


Figure 7.10: Read ACL on the Employee Training Records table

Access Control - u_employee_training_records

* Type: record Application: Global

* Operation: write Active: ☒

Admin overrides: ☒ Advanced: ☐

Protection policy: -- None --

* Name: Employee Training Records [u_employee_training_records] -- None --

Description: Allow write for records in u_employee_training_records, for users with role Hr Manager.

Conditions

Access Control Rules allow access to the specified resource if all four of these checks evaluate to true:

1. The user has one of the roles specified in the **Role** list, or the list is empty.
2. Users that do not satisfy the Security Attributes or the Subject Conditions will be denied access to records matching this ACL.
3. Conditions in the **Condition** field evaluate to true, or conditions are empty.
4. The script in the **Script** field (advanced) evaluates to true, or sets the variable "answer" to true, or is empty.

The four checks are evaluated independently in the order displayed above.

[More Info](#)

Requires role: 1 to 1 of 1

Figure 7.11: Write ACL on the Employee Training Records table

Employee Training Records Completion Date Search

Actions on selected rows... New

Completion Date	Employee	Status	Training Name
2024-10-27	Abel Tuter	Inprogress	full stack java
2024-12-31	Abel Tuter	Inprogress	Full Stack Java

1 to 2 of 2

Figure 7.12: Employee Training Records visible to the user with the Hr Manager role

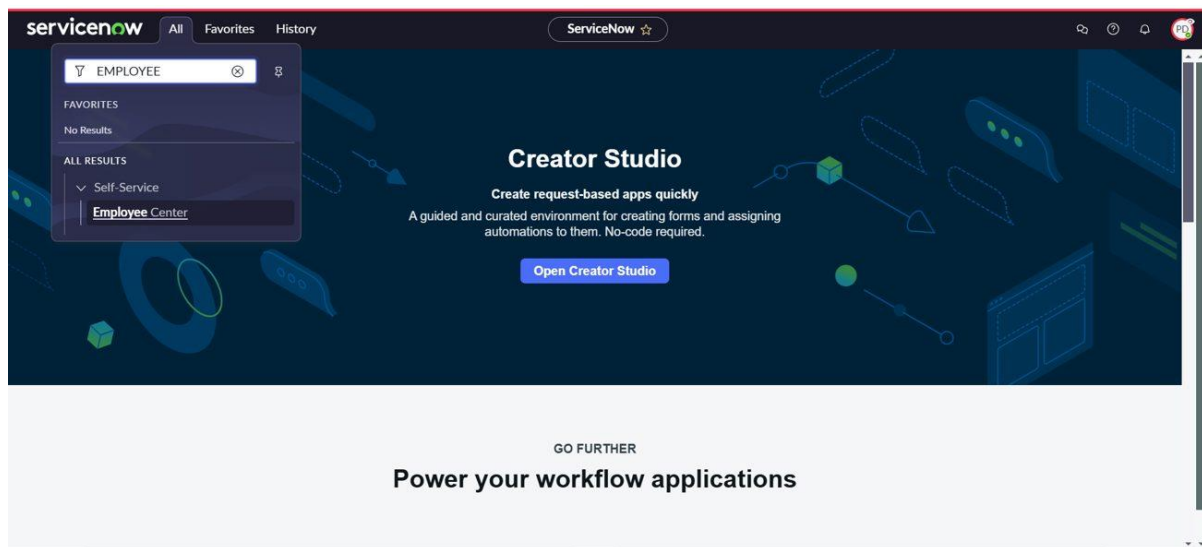


Figure 7.13: Other user cannot see the Employee Training Records table

8. Advantages & Disadvantages

Advantages

- Bulk data is loaded in minutes instead of being typed row by row.
- Staging table lets rows be checked before they reach the real table.
- Department comes from the employee record, so it stays correct without extra entry.
- Role-based ACLs protect the data on every request.
- No coding is needed; everything uses standard platform features.

Disadvantages

- Column names and choice values in the spreadsheet must match the table exactly, or rows fail or are skipped.
- Wrong or missing coalesce settings can create duplicate records.
- Reference fields need exact display values to be matched.
- ACLs need the security_admin role and careful testing.

9. Conclusion

This project demonstrated the essential process of importing data into ServiceNow using Import Sets and Transform Maps, using dot-walking to bring in related employee data, and applying ACLs to enforce data security. Together, these features gave accurate data integration, a clear link between each record and its employee, and protected sensitive information through role-based access control.

10. Future Scope

- Schedule recurring imports so new spreadsheets are loaded automatically.
- Set coalesce fields so repeated imports update existing records.
- Add transform scripts and choice actions to clean dates and status values.
- Build dashboards and reports on training status by department.
- Add notifications for pending or overdue training.

- Add field-level ACLs for more sensitive columns.

11. Appendix

Source Code (if any)

No custom source code was written. The project is configured with ServiceNow platform features (tables, import sets, transform maps, roles and ACLs).

Dataset Link

Dataset: Employee Training.xlsx (Training Name, Completion Date, Status, Employee). Link: [add your link here]

GitHub & Project Demo Link

GitHub: [add your link here] Project demo: [add your link here]